Lecture 9, Feb 9, 2026

* Indeterminate Products.

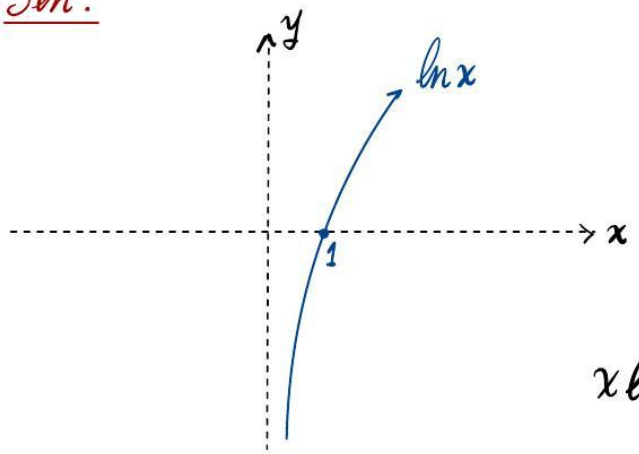
Let consider h, g such that $f(x) = h(x)g(x)$ Notice that.

$$f(x) = \frac{g(x)}{1/h(x)} = \frac{h(x)}{1/g(x)}$$

Indeterminate form of type $0 \cdot \infty$

Example. Evaluate $\lim_{x \to 0^+} x \ln x$.

Sln.



Notice that $\ln x \to -\infty$ as $x \to 0^+$

and $x \to 0$ as $x \to 0^+$

then we have $0 \cdot \infty$ then let

$x = 1/\frac{1}{x}$ then

$$x \ln x = \frac{\ln x}{1/x}$$

$$\lim_{x \to 0^+} x \ln x = \lim_{x \to 0^+} \frac{\ln x}{1/x}$$; $\ln x \to -\infty$ as $x \to 0^+$, $1/x \to \infty$ as $x \to 0^+$

Besides, $\frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2} \neq 0$ By L'Hospital's Rule.

$$\lim_{x \to 0^+} x \ln x \overset{L'H}{=} \lim_{x \to 0^+} \frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}(1/x)}$$

$$= \lim_{x \to 0^+} \frac{1/x}{-1/x^2} = \lim_{x \to 0^+} (-x) = 0$$

$$\lim_{x \to 0^+} x \ln x = \lim_{x \to 0^+} \frac{x}{1/\ln x}$$

Example. Use the L'Hôspital's rule to help sketch the graph of $f(x) = xe^x$.

Sln. Both x, e^x becomes large as $x \to \infty$ thus $\lim_{x\to\infty} xe^x = \infty$.

As $x \to -\infty$, but $e^x \to 0$. So,

$$\lim_{x\to-\infty} xe^x = \lim_{x\to-\infty} \frac{x}{e^{-x}}$$

$$\overset{L'H}{=} \lim_{x\to-\infty} \frac{1}{-e^{-x}} = -\lim_{x\to-\infty}(e^x) = 0.$$

thus the x-axis is a horizontal asymtote.

$$f'(x) = xe^x + e^x = (x+1)e^x.$$

$\mathrm{Dom} f = \mathbb{R}$. $f(x)$ is increasing for x in $(-1, \infty)$

$f(x)$ is decreasing for x in $(-\infty, -1)$.

$$f''(x) = (x+1)e^x + e^x = (x+2)e^x.$$

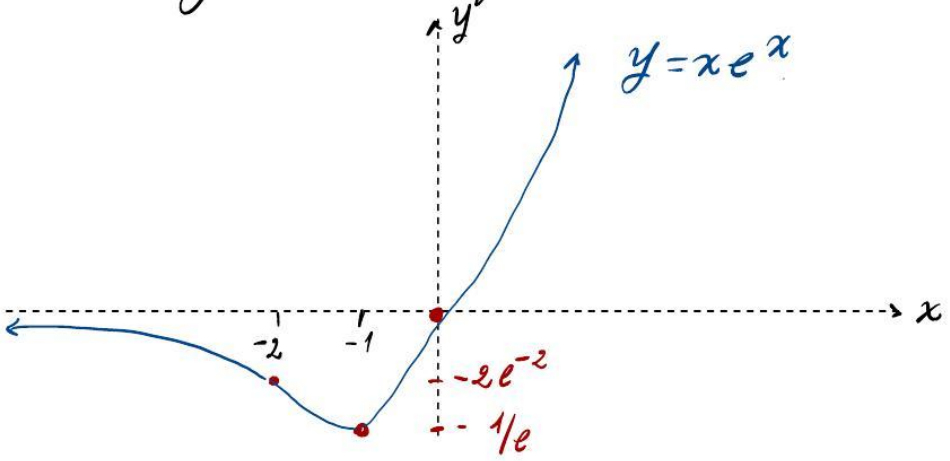
$f(x)$ is $\cup$ on $(-2, \infty)$ and f is $\cap$ on $(-\infty, -2)$.

$x = -1$ is the only critical point and

$f''(-1) = (-1+2)e^{-1} = e^{-1} > 0$ By the second derivative test at $x = -1$ f has a local minimum.

$f(-1) = -1e^{-1} = -\frac{1}{e}$ $P(-1, -\frac{1}{e})$ is a local minimum.

$x = -2$ is the only inflution point, i.e, $Q(-2, -2e^{-2})$



* Indeterminate Differences.

If $\lim_{x\to a} f(x) = \infty$ and $\lim_{x\to a} g(x) = \infty$ then the limit

$$\lim_{x\to a} [f(x) - g(x)]$$

is called an indeterminate form of type $\infty - \infty$.

Example. Compute $\lim_{x\to \frac{\pi}{2}^-} (\sec x - \tan x)$.

Sln Notice that $\sec x, \tan x \to \infty$ as $x \to \frac{\pi}{2}^-$. However.

$$\lim_{x\to \frac{\pi}{2}^-} (\sec x - \tan x) = \lim_{x\to \frac{\pi}{2}^-} \left(\frac{1}{\cos x} - \frac{\sin x}{\cos x}\right)$$

$$= \lim_{x\to \frac{\pi}{2}^-} \frac{1-\sin x}{\cos x}.$$

$$\overset{\frac{0}{0}}{\overset{L'H}{=}} \lim_{x\to \frac{\pi}{2}^-} \frac{-\cos x}{-\sin x} = 0.$$

* Indeterminate Powers.

Several indeterminate forms arise from the limit

$$\lim_{x\to a} [f(x)]^{g(x)}.$$

1). $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = 0$ type 0^0

2). $\lim_{x\to a} f(x) = \infty$ and $\lim_{x\to a} g(x) = 0$ type ∞^0

3). $\lim_{x\to a} f(x) = 1$ and $\lim_{x\to a} g(x) = \infty$ type 1^∞

For each of these three cases we can use

$$y = [f(x)]^{g(x)} \quad \text{then} \quad \ln y = g(x)\ln(f(x))$$

$$y = e^{\ln y} = e^{g(x)\cdot \ln(f(x))}.$$

Example. Calculate $\lim_{x\to 0^+} (1+\sin 4x)^{\cot x}$.

Sln. if $x\to 0^+$ then $(1+\sin 4x)\to 1$ and $\cot x \to \infty$. then we have 1^∞.

We know

$$y = e^{\cot x \cdot \ln(1+\sin 4x)},$$ since the function e^x is continuous then

$$\lim_{x\to 0^+} (1+\sin x)^{\cot x} = e^{\lim_{x\to 0^+} \cot x \cdot \ln(1+\sin 4x)}$$

$$\lim_{x\to 0} \cot x \cdot \ln(1+\sin 4x) = \lim_{x\to 0} \frac{\ln(1+\sin 4x)}{1/\cot x}$$

$$= \lim_{x\to 0^+} \frac{\ln(1+\sin 4x)}{\tan x} \quad \left(\frac{0}{0}\right).$$

By L'H $$= \lim_{x\to 0^+} \frac{\frac{4\cos 4x}{1+\sin 4x}}{\frac{1}{\cos^2 x}}$$

$$= \lim_{x\to 0^+} \frac{4\cos^2 x \cos 4x}{1+\sin 4x} = 4.$$

then $\lim_{x\to 0^+} (1+\sin 4x)^{\cot x} = e^4$

Example. Find $\lim_{x\to 0^+} x^x$

Sln.

$$x^x = e^{\ln(x^x)} = e^{x\ln x}.$$

$$\lim_{x\to 0^+} x^x = \lim_{x\to 0^+} e^{x\ln x} = e^{\lim_{x\to 0^+} x\ln x}$$

$\lim_{x \to 0^+} x \ln x = 0$. (By previous example).

Then $\lim_{x \to 0^+} x^x = e^0 = 1$.